01-teensy

File: sheets/01-teensy.kicad_sch

02-analog-front

File: sheets/02-analog-front.kicad_sch

03-adc

File: sheets/03-adc.kicad_sch

04-mcp-io

File: sheets/04-mcp-io.kicad_sch

05-cv-dac

File: sheets/05-cv-dac.kicad_sch

06-surface

File: sheets/06-surface.kicad_sch

07-midi

File: sheets/07-midi.kicad_sch

08-connectors-panel

File: sheets/08-connectors-panel.kicad_sch

09-connectors-rear

File: sheets/09-connectors-rear.kicad_sch

10-power

File: sheets/10-power.kicad_sch

Sheet: /
File: pcb-m.kicad_sch

Title:

Size: A4	Date:	**Rev:**
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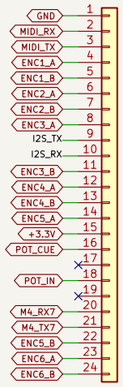
Teensy + Audio Shield: I2S interno al modulo; no microSD ni pot opcional del shield.

NC: T25/T27/T38/T39 reservados e-drum. HP_VGND nunca se une a GND/AGND.

Filas de sockets conservadas; validar mecanica del montaje antes de fabricar.

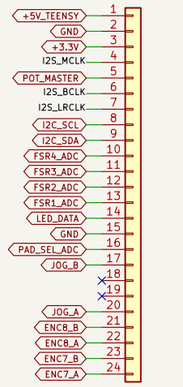
J_T41L

Teensy41-LeftRow



J_T41R

Teensy41-RightRow



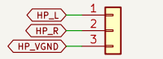
J_SHIELD_AUDIO

AudioShieldRevD-LineInOut



J_SHIELD_HP

AudioShieldRevD-HP



Sheet: /01-teensy/
File: 01-teensy.kicad_sch

Title:

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Frente analogico de la portadora Studio conservado (doc25 seccion 6).

D13: selector en panel: escalera 10k/10k a PAD_SEL_ADC. Puio C vía /_PAD_SEL (3 hilos).

SW_PAD representa polos A/B en panel, cableados al header 2x4c no montar switch en PCB-M.

R_PAD_L1
33k

R_PAD_L2
12k

R_PAD_L3
5.1k

SW_PADA
PAD 3P3T A/B

C_AC_L
1uF

R_SER_L
1k

R_BIAS_L
100k

D_CLAMP_L
BAT54S

JP_BYP_L
BYP L (sin U_OA_IN)

U_OA_INA
MCP6004

TP_BUF_L
BUF_L

C_LIN_L
4.7uF

R_PAD_R1
33k

R_PAD_R2
12k

R_PAD_R3
5.1k

SW_PADB
PAD 3P3T A/B

C_AC_R
1uF

R_SER_R
1k

R_BIAS_R
100k

D_CLAMP_R
BAT54S

JP_BYP_R
BYP R (sin U_OA_IN)

U_OA_INB
MCP6004

TP_BUF_R
BUF_R

C_LIN_R
4.7uF

R_VREF_T
10k

R_VREF_B
10k

C_VREF
10uF

U_OA_INC
MCP6004

JP_BYP_VREF
BYP VREF (sin U_OA_IN)

TP_VREF
VREF

U_OA_IND
MCP6004

U_OA_INE
MCP6004

C_OA_IN
100nF

C_LOUT_L
10uF

R_LOUT_L
47k

C_LOUT_R
10uF

R_LOUT_R
47k

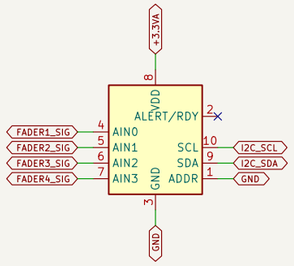
R_SEL_T
10k

R_SEL_B
10k

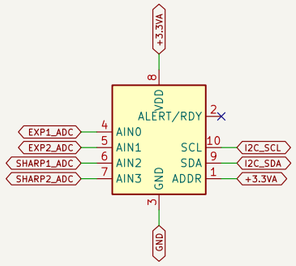
Sheet: /02-analog-front/		
File: 02-analog-front.kicad_sch		
Title:		
Size: A1	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 3/11

ADS1=0x48 (ADDR=GND); ADS2=0x49 (ADDR=VDD); ALERT/RDY no usado.

U_ADS1
ADS1115IDGSR



U_ADS2
ADS1115IDGSR



C_ADS1_VCC
100nF



C_ADS2_VCC
100nF



C_ADS1_BULK
10uF



C_ADS2_BULK
10uF



Sheet: /03-adc/		
File: 03-adc.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 4/11

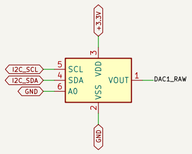
RESERVADO: MCP23017 0x23 vive en PCB-K. Hoja vacia conservada en la raiz.

Sheet: /04-mcp-io/	
File: 04-mcp-io.kicad_sch	
Title:	
Size: A3	Date:
KiCad E.D.A. 10.0.6	**Rev:** Id: 5/11

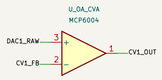
CV: 1 + 10k/20k = 1.5; A0 DAC1=GND (0x60), DAC2=3V3 (0x61).

D11: realimentacion antes de 1k serie: BAT54S en nodo protegido hacia jack, a GND/+5V_LED.

U_DAC1
MCP4725A0T-E/CH



U_OA_CVA
MCP6004

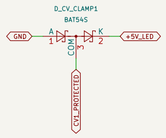


R_CV_RF1
10k

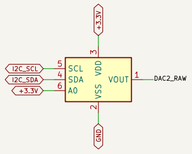
R_CV_RI1
20k

R_CV_SER1
1k

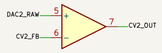
D_CV_CLAMP1
BAT54S



U_DAC2
MCP4725A0T-E/CH



U_OA_CVB
MCP6004

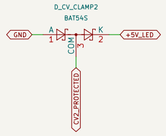


R_CV_RF2
10k

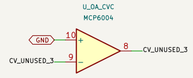
R_CV_RI2
20k

R_CV_SER2
1k

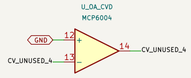
D_CV_CLAMP2
BAT54S



U_OA_CVC
MCP6004



U_OA_CVD
MCP6004



U_OA_CVE
MCP6004

C_DAC1_VCC
100nF



C_DAC2_VCC
100nF



C_OA_CV_VCC
100nF



C_DAC1_BULK
10uF



C_DAC2_BULK
10uF



C_OA_CV_BULK
10uF



Sheet: /05-cv-dac/	
File: 05-cv-dac.kicad_sch	
Title:	
Size: A2	Date:
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Componentes de panel por cable. JOG_CLICK via J_PCB_K hacia MCP 0x23.

D13: polo C comun a J_PAD_SEL:2; extremos a .1/.3; contacto central sin cable (1.65V).

D11: pots con RC 1k/100nF del lado ADC. J_HP unico jack HP; HP_VGND aislado de masa.

J_FSR1
JST-PH_2pin_FSR1
+3.3V
FSR1_ADC

J_FSR2
JST-PH_2pin_FSR2
+3.3V
FSR2_ADC

J_FSR3
JST-PH_2pin_FSR3
+3.3V
FSR3_ADC

J_FSR4
JST-PH_2pin_FSR4
+3.3V
FSR4_ADC

R_FSR1
10k
FSR1_ADC
GND

R_FSR2
10k
FSR2_ADC
GND

R_FSR3
10k
FSR3_ADC
GND

R_FSR4
10k
FSR4_ADC
GND

J_JOG
JOG_CABLE
GND
JOG_A
JOG_B
JOG_CLICK

J_POT_MASTER
POT_MASTER_CABLE
+3.3V
POT_MASTER_WIPER
GND

J_POT_CUE
POT_CUE_CABLE
+3.3V
POT_CUE_WIPER
GND

J_POT_IN
POT_IN_CABLE
+3.3V
POT_IN_WIPER
GND

J_PAD_SEL
PAD_SEL_CABLE
+3.3V
PAD_SEL_ADC
GND

J_HP
HP_CABLE
HP_L
HP_R
HP_VGND

R_POT_MASTER
1k
POT_MASTER_WIPER
POT_MASTER

R_POT_CUE
1k
POT_CUE_WIPER
POT_CUE

R_POT_IN
1k
POT_IN_WIPER
POT_IN

C_POT_MASTER
100nF
POT_MASTER
GND

C_POT_CUE
100nF
POT_CUE
GND

C_POT_IN
100nF
POT_IN
GND

Sheet: /06-surface/
File: 06-surface.kicad_sch
Title:
Size: A2
Date:
Rev:
KiCad E.D.A. 10.0.6
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MIDI TRS Type A: ring=fuente, tip=sumidero; entrada aislada, sleeve IN NC.

D12 CA-033 (2014): R_MIDI_R=33 ohm >=0.5W; R_MIDI_T=10 ohm >=0.25W; sin 100 ohm extra.

JST-PH hacia jacks de panel: 1=tip, 2=ring, 3=sleeve. Sin deteccion de insercion.

J_MIDI_IN
MIDI_IN_PANEL_CABLE
MIDI_IN_TIP
MIDI_IN_RING
1
2
3

U_OPT_MIDI
6N138
MIDI_OPTO_A
MIDI_IN_TIP
+5V_TEENSY
MIDI_OPTO_BASE
MIDI_RX
GND
1
2
3
4
5
6
7
8

R_MIDI_IN
220
MIDI_IN_RING
MIDI_OPTO_A

R_MIDI_RX
4.7k
+3.3V
MIDI_RX

C_MIDI_VCC
100nF
+5V_TEENSY
GND

R_MIDI_BASE
10k
MIDI_OPTO_BASE
GND

D_MIDI_IN
1N4148W
MIDI_OPTO_A
MIDI_IN_TIP

J_MIDI_OUT
MIDI_OUT_PANEL_CABLE
MIDI_OUT_TIP
MIDI_OUT_RING
GND
1
2
3

R_MIDI_T
10
MIDI_TX
MIDI_OUT_TIP

R_MIDI_R
33
+3.3V
MIDI_OUT_RING

Sheet: /07-midi/
File: 07-midi.kicad_sch

Title:

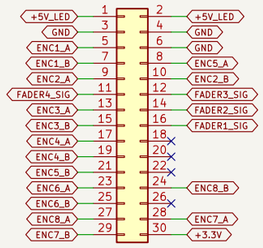
Size: A2	Date:	**Rev:**
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D8: J1 segun PCB-A fabricada: NC 18(spare)/20(INT_K)/22(INT_A)/26(LED_DIN).

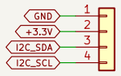
Encoders a Teensy segun firmware; faders al ADS0x48 AIN0-3. LED solo via J_PCB_K.

I2C hacia PCB-A por J2: J1 no lleva SDA/SCL. Retirar pull-ups PCB-A.

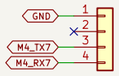
J1_PCBA
IDC_2x15_PCB-A



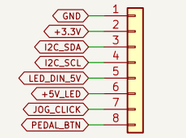
J2_I2C
JST-SH_4pin_I2C



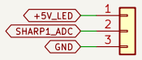
J3_M4
JST-PH_4pin_M4



J_PCB_K
PCB_K_KEY_CABLE



J5_SHARP1
JST-PH_3pin_Sharp1



J6_SHARP2
JST-PH_3pin_Sharp2



J7_EXP1
JST-PH_2pin_EXP1



J8_EXP2
JST-PH_2pin_EXP2



C_SHARP1
10uF



C_SHARP2
10uF



R_EXP1
10k

R_EXP2
10k

Sheet: /08-connectors-panel/
File: 08-connectors-panel.kicad_sch

Title:

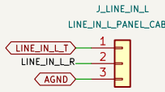
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 9/11

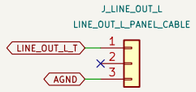
D15: Jacks con tuerca en pared trasera; JST-PH en borde PCB-M, uno por jack.

TRS: 1=tip/2=ring/3=sleeve; TS CV: 1=tip/2=sleeve. HP solo por J_HP (06).

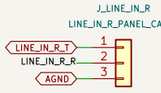
Captura anterior sin contactos de deteccion; no se agrega deteccion de insercion.

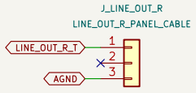
Anillos LINE IN expuestos como redes locales; terminacion balanceada sin definir.



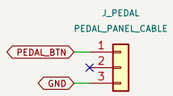














Sheet: /09-connectors-rear/		
File: 09-connectors-rear.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 10/11

LDO -> +3.3VA: +3.3V desde Teensy; +5V_LED separado de +5V_TEENSY. D10: NT1 junto a LDO. HP_VGND aislada.

D9: USB-A PI -> J_5V_LED_IN -> F1 0.75A -> +5V_LED. Backup DNP: sin TVS de entrada adecuada en BOM.

D14: pull-ups 2.2k en PCB-M: desoldar R1/R2 de PCB-A. Unica union AGND/GND por NT1.

F1
0.75A MF-MSMF075

U_LDO1
TLV75533PDBV

C_LDO_IN
10uF

C_LDO_OUT
1uF

U_LED_BUF
74AHCT1G125

J_5V_BKUP
5V_BACKUP_DNP

R_SDA
2.2k

R_SCL
2.2k

C_LED_BUF
100nF

J_5V_LED_IN
USB-A_PI_5V_LED_IN

NT1
Nettie_2

#FLG001
PWR_FLAG

#FLG002
PWR_FLAG

#FLG003
PWR_FLAG

#FLG004
PWR_FLAG

#FLG005
PWR_FLAG

#FLG006
PWR_FLAG

Sheet: /10-power/	
File: 10-power.kicad_sch	
Title:	
Size: A2	Date:
KiCad E.D.A. 10.0.6	**Rev:** Id: 11/11